

JESS Thermodynamic Database v8.9

Actinium

24-Jan-23

Reaction No. 2688							
F-1 + Ac+3 = Ac+3_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.591(4SF)	1	EOR	
2	25	0.1	NaClO4	lgK:2.95(3SF)	5	MTP	6681
3	25	0.5	NaClO4	lgK:2.72(3SF)	5	MDS	2106

Reaction No. 2689							
Ac+3_F-1 + F-1 = Ac+3_F-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:2.50(3SF)	5	MDS	2106

Reaction No. 2690							
Ac+3_F-1(2) + F-1 = Ac+3_F-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:2.68(3SF)	5	MDS	2106

Reaction No. 6330							
Oxalic-2 + Ac+3 = Ac+3_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.636(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:5.65(3SF)	4	CRV	31301
3	25	0.1	NaClO4	lgK:4.36(0.02SD)	5	MTP	906
4	25	0.5	NaClO4	lgK:4.47(3SF)	0	MDS	2106
5	25	1	NaClO4	lgK:3.56(3SF)	5	MPL	906
6	25	1	NaClO4	lgK:3.56(3SF)	6	MPL	999

Reaction No. 6331							
2<Oxalic-2> + Ac+3 = Ac+3_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.777(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:8.80(3SF)	4	CRV	31301

3	25	0.1	NaClO ₄	lgK:7.1(0.1SD)	4	MTP	906
4	25	0.1	NaClO ₄	lgK:7.08(0.1SD)	4	MTP	906
5	25	0.5	NaClO ₄	lgK:8.00(3SF)	0	MDS	2106
6	25	1	NaClO ₄	lgK:6.16(3SF)	5	MPL	906
7	25	1	NaClO ₄	lgK:6.16(3SF)	6	MPL	999

Reaction No. 43706							
Ac+3_OH-1 + Citric-3 = Ac+3_OH-1_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:4.18(3SF)	4	ENS	906

Reaction No. 43707							
Ac+3_OH-1 + 2<Citric-3> = Ac+3_OH-1_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:7.0(2SF)	4	ENS	906

Reaction No. 43708							
Ac+3_OH-1(2) + Citric-3 = Ac+3_OH-1(2)_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:4.3(2SF)	4	ENS	906

Reaction No. 44822							
EDTA-4 + Ac+3 = Ac+3_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:14.2(0.1SD)	2	MTP	906
2	25	0	Inf. Dilution	lgK:14.4(3SF)	1	ESC	

Reaction No. 44823							
Ac+3_EDTA-4 + H+1 = Ac+3_H+1_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	Unknown	lgK:2.(0.3SD)	4	MTP	906
2	25	0	Inf. Dilution	lgK:2.0(2SF)	1	ESC	

Reaction No. 45241							
CDTA-4 + Ac+3 = Ac+3_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:15.7(0.1SD)	4	MTP	906

Reaction No. 52987							
Cl-1 + Ac+3 = Ac+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3	LiClO4	lgK:-0.36 (2SF)	5	CRV	2556
2	25	0	Inf. Dilution	lgK:0.9706 (4SF)	1	EOR	
3	25	4	NaClO4	lgK:-0.04 (1SF)	5	MDS	5398
4	27	1	HClO4	lgK:-0.10 (2SF)	5	MDS	931

Reaction No. 52988							
2<Cl-1> + Ac+3 = Ac+3_Cl-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.403 (4SF)	1	EOR	
2	25	4	NaClO4	lgK:-1.04 (3SF)	5	MDS	5398
3	27	1	HClO4	lgK:-0.62 (2SF)	5	MDS	931

Reaction No. 52989							
3<Cl-1> + Ac+3 = Ac+3_Cl-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.809 (4SF)	1	EOR	
2	25	4	NaClO4	lgK:-1.26 (3SF)	5	MDS	5398

Reaction No. 53298							
SO4-2 + Ac+3 = Ac+3_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.341 (4SF)	1	EOR	
2	25	0.5	NaClO4	lgK:1.75 (3SF)	5	MDS	5398
3	25	1	NaClO4	lgK:1.36 (3SF)	5	MDS	5398
4	27	1	HClO4	lgK:1.20 (3SF)	5	MDS	931

Reaction No. 53299							
2<SO4-2> + Ac+3 = Ac+3_SO4-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.041 (4SF)	1	EOR	
2	25	0.5	NaClO4	lgK:2.64 (3SF)	5	MDS	5398
3	25	1	NaClO4	lgK:2.68 (3SF)	5	MDS	5398
4	27	1	HClO4	lgK:1.85 (3SF)	0	MDS	931

Reaction No. 53369							
$\text{Ac}+3 + \text{H}+1(2)\text{PO}_4-3 = \text{Ac}+3\text{H}+1(2)\text{PO}_4-3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NH ₄ ClO ₄	lgK:1.59(3SF)	5	MDS	5398

Reaction No. 53836							
$\text{NO}_3-1 + \text{Ac}+3 = \text{Ac}+3\text{NO}_3-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	27	1	HClO ₄	lgK:0.12(2SF)	5	MDS	931

Reaction No. 53837							
$2<\text{NO}_3-1> + \text{Ac}+3 = \text{Ac}+3\text{NO}_3-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	27	1	HClO ₄	lgK:0.01(1SF)	5	MDS	931

Reaction No. 54283							
$\text{Br}-1 + \text{Ac}+3 = \text{Ac}+3\text{Br}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3	LiClO ₄	lgK:-0.4(1SF)	5	CRV	2556
2	27	1	HClO ₄	lgK:-0.25(2SF)	4	MDS	931

Reaction No. 54284							
$2<\text{Br}-1> + \text{Ac}+3 = \text{Ac}+3\text{Br}-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	27	1	HClO ₄	lgK:-0.52(2SF)	4	MDS	931

Reaction No. 55651							
$\text{Ac}+3 + \text{e}-1 = \text{Ac}+2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-82.9(3SF)	1	ELC	
2	25	0	Unknown	E:-5.(0.5SD)	0	ENS	5256

Reaction No. 57706							
$\text{Ac}(s) - 3<\text{e}-1> = \text{Ac}+3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-150.(3.SD)	0	ENS	3006
2	25	0	Inf. Dilution	dG:-640.(3SF)kJ	3	CRV	7421

3	25	0	Inf. Dilution	E:2.20 (3SF)	2	CRV	6330
4	25	0	Inf. Dilution	E:2.13 (3SF)	0	ENS	7276
5	25	0	Inf. Dilution	E:2.20 (3SF)	2	CRV	7761
6	25	0	Inf. Dilution	E:2.20 (3SF)	2	CRV	22519
7	25	0	Inf. Dilution	E:2.15 (3SF)	0	CRV	22551
8	25	0	Inf. Dilution	dH:-160. (3.SD)	1	ENS	3006
9	25	0	Inf. Dilution	dH:-653.2 (4SF) kJ	1	CRV	7761

Reaction No. 60666							
2<F-1> + Ac+3 = Ac+3_F-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.699 (4SF)	1	EOR	
2	25	0.5	Unknown	lgK:5.22 (3SF)	5	CRV	2556

Reaction No. 60667							
3<F-1> + Ac+3 = Ac+3_F-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.675 (4SF)	1	EOR	
2	25	0.5	Unknown	lgK:7.9 (2SF)	5	CRV	2556

Reaction No. 65703							
SCN-1 + Ac+3 = Ac+3_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.388 (4SF)	1	EOR	
2	25	1	Unknown	lgK:0.05 (1SF)	6	CRV	552
3	25	5	NaClO4	lgK:-0.75 (2SF)	5	MDS	11516
Note (3): Sekine T et al., Bull. Chem. Soc. Jpn., 1969, 42, 2712							

Reaction No. 70522							
Ac(s) = Ac(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:366.0 (4SF) kJ	0	CRV	6330
2	25	0	Inf. Dilution	dG:87.6 (3SF)	4	CRV	7382
3	25	0	Inf. Dilution	dH:406.0 (4SF) kJ	0	CRV	6330
4	25	0	Inf. Dilution	dH:97.0 (3SF)	4	CRV	7382
5	25	0	Inf. Dilution	dH:410. (3SF) kJ	3	CRV	22524

Reaction No. 72149							
$\text{Ac}+3 + 1.5\text{H}_2(\text{g}) + 3\text{e}^- = \text{AcH}_3(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.70 (3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:452.4 (4SF) kJ	3	CRV	7761

Reaction No. 72150							
$\text{Ac}+3_{\text{OH}-1(3)}(\text{am},\text{s}) + 3\text{e}^- = \text{Ac}(\text{s}) + 3\text{OH}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.53 (3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:650. (3SF) kJ	3	EES	
Note (2): By analogy with the crystalline solid							

Reaction No. 72151							
$\text{Ac}+3_{\text{OH}-1(3)}(\text{s}) + 3\text{e}^- = \text{Ac}(\text{s}) + 3\text{OH}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-132.7 (4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.57 (3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:656.7 (4SF) kJ	2	CRV	7761

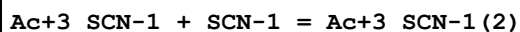
Reaction No. 73882							
$\text{Ac}(\text{s}) - 2\text{e}^- = \text{Ac}+2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.3 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-167. (3SF) kJ	2	CRV	22524

Reaction No. 74042							
$\text{Ac}+3_{\text{OH}-1(3)}(\text{s}) = \text{Ac}+3 + 3\text{OH}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-20.9 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-20.9 (3SF)	4	CRV	22551

Reaction No. 75930							
$\text{Ac}+3_{\text{Oxalic}-2} + \text{Oxalic}-2 = \text{Ac}+3_{\text{Oxalic}-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO ₄	lgK:3.53 (3SF)	0	MDS	2106
2	25	1	NaClO ₄	lgK:2.60 (3SF)	5	MDS	11516

Note (2): Sekine T et al., Bull. Chem. Soc. Jpn., 1969, 42, 2712

Reaction No. 76130



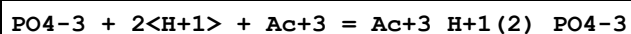
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	5	NaClO4	lgK:0.29(2SF)	5	MDS	11516

Note (1): Sekine T et al., Bull. Chem. Soc. Jpn., 1969, 42, 2712

2	27	1	NaClO4	lgK:-0.14(2SF)	5	MDS	11516
---	----	---	--------	----------------	---	-----	-------

Note (2): Rao C et al., Inorg. Nucl. Chem. Lett., 1968, 4, 655

Reaction No. 77976



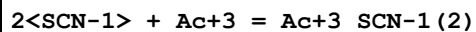
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.0(4SF)	1	ELC	

Reaction No. 79195



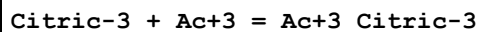
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.5(3SF)	1	ELC	

Reaction No. 79440



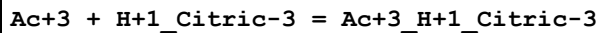
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.904(4SF)	1	ELC	

Reaction No. 80633



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.99(3SF)	4	CRV	31301

Reaction No. 80634



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.31(3SF)	4	CRV	31301

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EES	Personal estimate
ELC	Linear Combination of Reactions
ENS	Not specified
EOR	Other Reaction Constants
ESC	Standard conditions anchor
MDS	Distribution between two phases
MPL	Polarography
MTP	Migration or transference number

References

Reference No.: 552(95SMM)

NIST Critical Stability Constants of Metal Complexes Database, 1995 Smith RM; Martell AE; Motekaitis RJ; Version 2.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 906(79Per)

IUPAC Chemical Data Series No. 22, 1979 Perrin DD; Stability Constants of Metal-Ion Complexes, First Edition Part B, Organic Ligands; Pergamon, Oxford

Reference No.: 931(71SiM)

Stability Constants of Metal-ion Complexes SP25, 1971 Sillen LG; Martell AE; Supplement No 1 to Special Publication No 25; Chemical Society, London

Reference No.: 999(91MaM)

Estimation of Constants for JESS Database, 1991 May PM; May RF

Reference No.: 2106(70AzL)

J. Inorg. Nucl. Chem., 1970, 32, 1925 - 1932; Aziz A; Lyle SJ; Complexes of Lanthanum and Actinium with Fluoride, Oxalate and Sulphate in Aqueous Solutions

Reference No.: 2556(93MSM)

NIST Critical Stability Constants of Metal Complexes Database, 1993 Martell AE; Smith RM; Motekaitis RJ; Version 1.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 3006(85BPJ)

Standard Potentials in Aqueous Solution, 1985 Bard AJ; Parsons R; Jordan J; Monographs in Electroanalytical Chemistry and Electrochemistry; Marcel Dekker Inc., New York and Basel

Reference No.: 5256(91MiK)

Coord. Chem. Rev., 1991, 109, 1 - 5; Mikheev NB; Kamenskaya AN; Complex Formation of the Lanthanides and Actinides in Lower Oxidation States

Reference No.: 5398(82Hog)

IUPAC Chemical Data Series No. 21, 1982 Hogfeldt E; Stability Constants of Metal-ion Complexes, 2nd Suppl., Part A, Inorganic Ligands; Pergamon, Oxford

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 6681(73MSS)

Russ. J. Inorg. Chem., 1973, 18, 783 - 785; Makarova TP; Stepanov AV; Shestakov BI; Electromigration Investigation of the Comparative Stability of Fluoro-complexes of the MF₂ Type of Certain Rare-Earth and Actinide Elements

Reference No.: 7276(88CoW)

Advanced Inorganic Chemistry, 5th Edn., 1988 Cotton FA; Wilkinson G; John Wiley & Sons Inc., N.Y., U.S.A

Reference No.: 7382(86Wea)

CRC Handbook of Chemistry and Physics, 67th Edn., 1986 Weast RC; A Ready-Reference Book of Chemical and Physical Data; CRC Press Inc., Boca Raton, Florida, U.S.A.

Reference No.: 7421(97Mar)

Ion Properties, 1997 Marcus Y; Marcel Dekker Inc., New York, U.S.A.

Reference No.: 7761(89Bra)

J. Phys. Chem. Ref. Data, 1989, 18, 1 - 21; Bratsch SG; Standard Electrode Potentials and Temperature Coefficients in Water at 298.15 K

Reference No.: 11516(00PeP)

IUPAC Stability Constants Database, 2000 Pettit LD; Powell HKJ; Release 4; Academic Software, U.K.

Reference No.: 22519(05Lid)

CRC Handbook of Chemistry and Physics, 86th Edn., 2005 Lide DR; A Ready-Reference Book of Chemical and Physical Data; Taylor & Francis, Boca Raton, USA

Reference No.: 22524(86BrL)

J. Phys. Chem., 1986, 90, 307 - 312; Bratsch SG; Lagowski JJ; Actinide Thermodynamic Predictions. 3. Thermodynamics of Compounds and Aquo Ions of the 2+, 3+, and 4+ Oxidation States and Standard Electrode Potentials at 298.15 K

Reference No.: 22551(06Inz)

Encyclopedia of Electrochemistry, Vol. 7a, 2006, 17 - 75; Inzelt G; Standard, Formal and Other Characteristic Potentials of Selected Electrode Reactions

Reference No.: 31301(10KFD)

Japan Atomic Energy Agency Data/Code 2009-024, 2010 Kitamura A; Fujiwara K; Doi R; Yoshida Y; Mihara M; Terashima M; Yui M; JAEA Thermodynamic Database for Performance Assessment of Geological Disposal of High-level Radioactive and TRU Wastes; Geological Isolation Research Unit, JAEA, Japan